Week 4

1-DP-Playing with Numbers

```
#include <stdio.h>

int main() {
    int n, i;
    long long dp[1000];

    scanf("%d", &n);

    dp[0] = 1;

    for (i = 1; i <= n; i++) {
        dp[i] = dp[i - 1];

        if (i >= 3)
            dp[i] += dp[i - 3];
    }

    printf("%lld", dp[n]);

    return 0;
}
```

2-DP-Playing with Chessboard

```
#include <stdio.h>

int main() {
    int n, i, j;
    int a[100][100], dp[100][100];

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &a[i][j]);

    dp[0][0] = a[0][0];

    for (i = 1; i < n; i++)
        dp[i][0] = dp[i - 1][0] + a[i][0];

    for (j = 1; j < n; j++)
        dp[0][j] = dp[0][j - 1] + a[0][j];

    for (i = 1; i < n; i++) {
        for (j = 1; j < n; j++) {
            if (dp[i - 1][j] > dp[i][j - 1])
                dp[i][j] = dp[i - 1][j] + a[i][j];
            else
                dp[i][j] = dp[i][j - 1] + a[i][j];
        }
    }

    printf("%d", dp[n - 1][n - 1]);

    return 0;
}
```

3-DP-Longest Common Subsequence

```
#include <stdio.h>
#include <string.h>

int main() {
    char s1[100], s2[100];
    int dp[100][100];
    int i, j, m, n;

    scanf("%s", s1);
    scanf("%s", s2);
    m = strlen(s1);
```

```
    n = strlen(s2);

    for (i = 0; i <= m; i++)
        dp[i][0] = 0;

    for (j = 0; j <= n; j++)
        dp[0][j] = 0;

    for (i = 1; i <= m; i++) {
        for (j = 1; j <= n; j++) {
            if (s1[i - 1] == s2[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else if (dp[i - 1][j] > dp[i][j - 1])
                dp[i][j] = dp[i - 1][j];
            else
                dp[i][j] = dp[i][j - 1];
        }
    }

    printf("%d", dp[m][n]);

    return 0;
}
```

4-DP-Longest non-decreasing Subsequence

```
#include <stdio.h>

int main() {
    int n, a[100], dp[100];
    int i, j, max = 1;

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++)
        dp[i] = 1;

    for (i = 1; i < n; i++) {
        for (j = 0; j < i; j++) {
            if (a[j] <= a[i] && dp[j] + 1 > dp[i])
                dp[i] = dp[j] + 1;
        }

        if (dp[i] > max)
            max = dp[i];
    }

    printf("%d", max);

    return 0;
}
```